

Volkan Ozten

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Skills:

- Proficient with C, C++, Java, Python, MATLAB, RISC-V Assembly, and Verilog/SystemVerilog HDL
- Proficient with various ECE tools such as DMMs, Waveform generators, soldering, and oscilloscopes
- Proficient with Git and GitHub for version control, branching workflows, and collaborative development
- Proficient with Vivado Design Suite and Vitis Software Platform
- Experienced in using FPGA development boards such as Digilent Nexys A7-100T
- Proficient with MS Windows, UNIX (Linux) Operating systems, and their respective terminal workflows
- Experienced in using Microsoft Office, Excel, PowerPoint, and Visio

Experience:**Consolidated Ocean Technologies, Inc. – Electrical Engineering Intern**

July 2023 – Present

- Selected and integrated an appropriate remote controller for an autonomous underwater vehicle (AUV) through hardware research and performance evaluation; developed C# interface code to communicate with the main control algorithm.
- Diagnosed and resolved RS-232 protocol-level issues between embedded systems on an AUV and a GPS flasher unit, enabling robust bidirectional communication.
- Engineered and documented AUV electrical cabling architecture, optimizing wire gauge, connector types, and routing paths for minimal noise, weight, and assembly complexity.
- Drafted electrical drawings for battery packs, power distribution systems, and other electrical systems for use in an AUV utilizing PowerPoint and Visio.
- Assist with the development and debugging of mechanical, electrical, and software interfaces of a control system housing an inertial navigation system for an AUV.

Capstone Design Project – Handheld Thermal Imaging System w/ External Calibration Protocol

UC Santa Barbara | Partnered with Teledyne FLIR, September 2024 – June 2025

- Evaluated and selected embedded processing platforms, including Broadcom BCM2710A1 (Raspberry Pi Zero 2 W) and BCM2711 (Raspberry Pi 4 Model B), balancing system constraints of size, power, and peripheral support.
- Designed and implemented the full software control flow in C++ and Python, integrating thermal camera input, image processing, display output, and user interface logic across heterogeneous embedded processors (Broadcom BCM2710A1 and BCM2711).
- Collaborated on electrical and mechanical integration of the handheld thermal system, conducting power characterization, sensor calibration, and enclosure thermography to validate system reliability and performance under operational heat and load profiles.
- Awarded **Best Multidisciplinary Project** among all UCSB 2025 Engineering Capstone teams.

Sonatech, LLC – Assistant Test Technician

July 2023 – September 2023

- Conducted electronic acceptance tests for transponders utilizing MATLAB and LabVIEW to test if transponders were performing their intended function.
- Improved reliability of transponder signal verification by expanding and optimizing MATLAB testing procedures for fidelity and performance validation.
- Diagnosed and corrected issues with electrical acceptance testing procedures and their respective devices (transponders).

Education:**University of California, Santa Barbara**

M.S. in Computer Engineering – Expected June 2026

Focus: Embedded Systems, Computer Architecture, and VLSI

B.S. in Electrical Engineering – Graduated June 2025, GPA of 3.66

Honors: Dean's List, **Best Multidisciplinary Capstone Project**

Extracurriculars/Volunteer Experience:

- UCSB IEEE Member
- Tutor for AP Physics 1 & 2 in High School
- Varsity Swim Team and Water Polo in High School